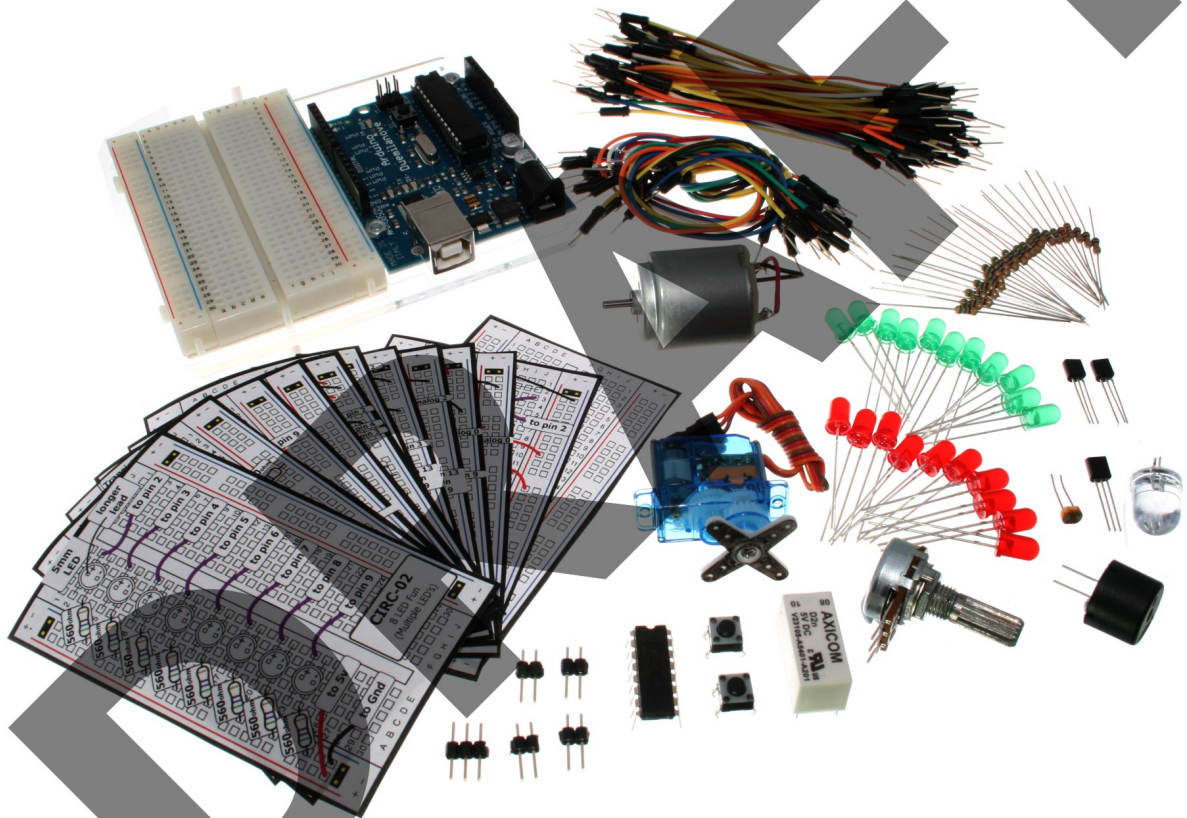


Experimenter's Guide for Arduino



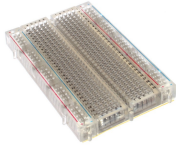
DESIGN REFERENCE



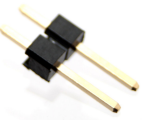
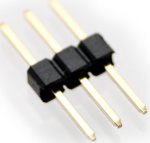
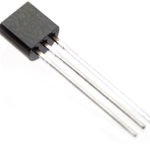
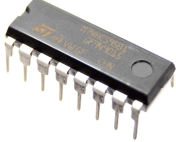
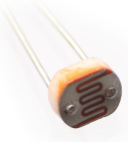
.:Part Summary:.

ID	Name	Qty	Description	Datasheet
ARDH-01	Arduino Holder	1	Acrylic Arduino Holder (109mm x 94mm)	
ARDU-01-C	Arduino	1	Arduino Duemilanove 328	
BATT-09	Battery Clip	1	Plastic Battery Snap (PP9)	http://oomlout.com/B02-D
BOL-03-10	3mmx10mm Machine Screw	2	3mm x 10mm Machine Screw (zinc plated)	
BREA-01-T	Breadboard	1	400 Point breadboard (82x53x8.5mm)	
BUMP-01	Bumpers	4	Rubber Bumper (Transparent 9.5mm x 3.8 mm)	http://oomlout.com/B2B-D
BUTT-01	Pushbutton	3	6mm x 6mm Tactile Through Hole Pushbutton	http://oomlout.com/B2C-D
CAPA-220	220uF Capacitor	1	220uF Capacitor (16v 2.5mm Lead Spacing)	http://oomlout.com/B36-D
DIOD-01	Diode	2	1N4001 Diode (1 amp)	http://oomlout.com/B45-D
HEAD-02	2 Pin Header	8	2 Pin Header (2.54 mm Normal Length)	
HEAD-03	3 Pin Header (Long)	1	3 Pin Header (2.54 mm Long)	
IC-2222A	Transistor	3	NPN Transistor (600 mA CBE)	http://oomlout.com/B79-D
IC-74HC595	Shift Register	1	8 Bit Serial In Parallel Out Shift Register With Latch	http://oomlout.com/B85-D
IC-PHOTO-01	Photo Resistor	1	Light Dependant Resistor (5mm Lead Spacing)	http://oomlout.com/B95-D
IC-TMP36	Temperature Sensor	1	Analog Temperature Sensor (-40-120 degrees)	http://oomlout.com/B9D-D
JUMP-01-C	Jumper Wires	1	70 piece Jumper Wire Bundle	
LED-05G	Green LED	12	5mm Green LED	http://oomlout.com/BC1-D
LED-05R	Red LED	12	5mm Red LED	http://oomlout.com/BC7-D
LED-10B	Big LED	1	10mm Clear Lense LED	http://oomlout.com/BD0-D
MOTOR-01	DC Motor	1	Small Toy Motor	http://oomlout.com/BDE-D
NUT-03-01	3mm Nut	4	3mm Nut Zinc Plated Full	
PBOX-01	Box	1	Plastic Box (23.5 cm x 13 cm x 3 cm)	
PIEZ-01	Piezo	1	Piezo Element 70DB 14mm 2KHz	http://oomlout.com/BFA-D
PLUG-01	DC Plug	1	2.1mm DC Battery Plug	http://oomlout.com/BFB-D
POT-103-A	Potentiometer	1	Breadboard Com	http://oomlout.com/C05-D
RELAY-01	Relay	1	PCB Mount Double Pole Double Throw Relay (5 volts)	http://oomlout.com/C0A-D
RES-103	10k Ohm Resistor	5	Resistor ¼ watt 10k Ohm (Brown-Black-Orange)	
RES-222	2.2k Ohm Resistor	5	Resistor ¼ watt 2.2k Ohm (Red-Red-Red)	
RES-561	560 Ohm Resistor	30	Resistor ¼ watt 560 Ohm (Blue-Green-Brown)	
SERV-04-A	Servo	1	Mini Servo (Spring RC SM2309-S)	http://oomlout.com/C3A-D
USBC-01-C	USB Cable	1	USB Cable 2m A to B	

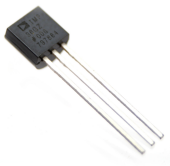
..Part Summary (detail):. (1 of 5)

	ID: ARDH-01 QTY 1 Manufacturer: Name: Arduino Holder Description: Acrylic Arduino Holder (109mm x 94mm) Datasheet: Source: <i>Farnell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i>
	ID: ARDU-01-C QTY 1 Manufacturer: Name: Arduino Description: Arduino Duemilanove 328 Datasheet: Source: <i>Farnell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> 1813412 696-1655
	ID: BATT-09 QTY 1 Manufacturer: Name: Battery Clip Description: Plastic Battery Snap (PP9) Datasheet: http://www.oomlout.com/B02-D Source: <i>Farnell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> 1650667 489-021 563-HH-3449 377-1549-ND
	ID: BOL-03-10 QTY 2 Manufacturer: Name: 3mmx10mm Machine Screw Description: 3mm x 10mm Machine Screw (zinc plated) Datasheet: Source: <i>Farnell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> 1419292 560596
	ID: BREA-01-T QTY 1 Manufacturer: BB-801-T Name: Breadboard Description: 400 Point breadboard (82x53x8.5mm) Datasheet: Source: <i>Farnell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> MOQ: 500
	ID: BUMP-01 QTY 4 Manufacturer: 3M SJ-5306 Name: Bumpers Description: Rubber Bumper (Transparent 9.5mm x 3.8 mm) Datasheet: http://www.oomlout.com/B2B-D Source: <i>Farnell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> 517-SJ-5306CL
	ID: BUTT-01 QTY 3 Manufacturer: FSM4JH TYCO Name: Pushbutton Description: 6mm x 6mm Tactile Through Hole Pushbutton Datasheet: http://www.oomlout.com/B2C-D Source: <i>Farnell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> 1555982 479-1413 450-1650-ND

..Part Summary (detail):. (2 of 5)

	ID: CAPA-220 QTY 1 Manufacturer: Name: 220uF Capacitor Description: 220uF Capacitor (16v 2.5mm Lead Spacing) Datasheet: http://www.oomlout.com/B36-D Source: <i>Farnell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> 9451099 5478161
	ID: DIOD-01 QTY 2 Manufacturer: Name: Diode Description: 1N4001 Diode (1 amp) Datasheet: http://www.oomlout.com/B45-D Source: <i>Farnell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> 1562268 628-8931 512-1N4001 641-1310-1-ND
	ID: HEAD-02 QTY 8 Manufacturer: TruConnect Name: 2 Pin Header Description: 2 Pin Header (2.54 mm Normal Length) Datasheet: Source: <i>Farnell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> 1097954 156049
	ID: HEAD-03 QTY 1 Manufacturer: Name: 3 Pin Header (Long) Description: 3 Pin Header (2.54 mm Long) Datasheet: Source: <i>Farnell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> 1097954 156049
	ID: IC-2222A QTY 3 Manufacturer: P2N2222AG ON Semi Name: Transistor Description: NPN Transistor (600 mA CBE) Datasheet: http://www.oomlout.com/B79-D Source: <i>Farnell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> 1611371 5449624 863P2N2222AG
	ID: IC-74HC595 QTY 1 Manufacturer: 74HC595N (NXP) Name: Shift Register Description: 8 Bit Serial In Parallel Out Shift Register With Latch (74HC595) Datasheet: http://www.oomlout.com/B85-D Source: <i>Farnell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> 3166028 4836645 511-M74HC595 568-1484-5-ND
	ID: IC-PHOTO-01 QTY 1 Manufacturer: Vactec VT935G Name: Photo Resistor Description: Light Dependant Resistor (5mm Lead Spacing) Datasheet: http://www.oomlout.com/B95-D Source: <i>Farnell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> 1652638 4558036

..Part Summary (detail):. (3 of 5)

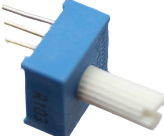
	ID: IC-TMP36 QTY 1 Manufacturer: TMP36TMP36GT9Z Analog Name: Temperature Sensor Description: Analog Temperature Sensor (-40-120 degrees) Datasheet: http://www.oomlout.com/B9D-D Source: <i>Famell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> 1438760 427351 TMP36GT9Z-ND
	ID: JUMP-01-C QTY 1 Manufacturer: BBJ70 Name: Jumper Wires Description: 70 piece Jumper Wire Bundle Datasheet: Source: <i>Famell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> MOQ: 500
	ID: LED-05G QTY 12 Manufacturer: FLR-50A04-PG5 Name: Green LED Description: 5mm Green LED Datasheet: http://www.oomlout.com/BC1-D Source: <i>Famell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> 1581138 2471678 859LTL4234
	ID: LED-05R QTY 12 Manufacturer: FLR-50A03-SR7 Name: Red LED Description: 5mm Red LED Datasheet: http://www.oomlout.com/BC7-D Source: <i>Famell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> 1581135 6194410 78TLHR5400
	ID: LED-10B QTY 1 Manufacturer: Name: Big LED Description: 10mm Clear Lense LED Datasheet: http://www.oomlout.com/BD0-D Source: <i>Famell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> 1168543 2471842
	ID: MOTOR-01 QTY 1 Manufacturer: Name: DC Motor Description: Small Toy Motor Datasheet: http://www.oomlout.com/BDE-D Source: <i>Famell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> 599116
	ID: NUT-03-01 QTY 4 Manufacturer: Name: 3mm Nut Description: 3mm Nut Zinc Plated Full Datasheet: Source: <i>Famell</i> <i>RS</i> <i>Mouser</i> <i>DigiKey</i> <i>Larger Volumes (contact oomlout)</i> 1419447 560293

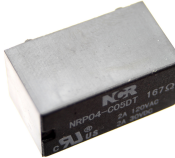
..Part Summary (detail):. (4 of 5)

	ID:	PBOX-01	QTY	1	
	Manufacturer:	K042022S			
	Name:	Box			
	Description:	Plastic Box (23.5 cm x 13 cm x 3 cm)			
	Datasheet:				
	Source:	Farnell	RS	Mouser	DigiKey
					Larger Volumes (contact oomlout)
					MOQ: 1000

	ID:	PIEZ-01	QTY	1	
	Manufacturer:	PS1420P02AT TDK			
	Name:	Piezo			
	Description:	Piezo Element 70DB 14mm 2KHz			
	Datasheet:	http://www.oomlout.com/BFA-D			
	Source:	Farnell	RS	Mouser	DigiKey
		1669967		810PS1420P02AT	445-2526-3-ND
					Larger Volumes (contact oomlout)

	ID:	PLUG-01	QTY	1	
	Manufacturer:				
	Name:	DC Plug			
	Description:	2.1mm DC Battery Plug			
	Datasheet:	http://www.oomlout.com/BFB-D			
	Source:	Farnell	RS	Mouser	DigiKey
		224923	455-132	693-4840.1201	CP3-1000-ND
					Larger Volumes (contact oomlout)

	ID:	POT-103-A	QTY	1	
	Manufacturer:	3386T			
	Name:	Potentiometer			
	Description:	Breadboard Compatible 10 k Ohm Potentiometer			
	Datasheet:	http://www.oomlout.com/C05-D			
	Source:	Farnell	RS	Mouser	DigiKey
					Larger Volumes (contact oomlout)
					MOQ: 10000

	ID:	RELAY-01	QTY	1	
	Manufacturer:	13937936 Tyco			
	Name:	Relay			
	Description:	PCB Mount Double Pole Double Throw Relay (5 volts)			
	Datasheet:	http://www.oomlout.com/C0A-D			
	Source:	Farnell	RS	Mouser	DigiKey
		4219582			Larger Volumes (contact oomlout)

	ID:	RES-103	QTY	5	
	Manufacturer:				
	Name:	10k Ohm Resistor			
	Description:	Resistor ¼ watt 10k Ohm (Brown-Black-Orange)			
	Datasheet:				
	Source:	Farnell	RS	Mouser	DigiKey
		9339060	135910		Larger Volumes (contact oomlout)

	ID:	RES-222	QTY	5	
	Manufacturer:				
	Name:	2.2k Ohm Resistor			
	Description:	Resistor ¼ watt 2.2k Ohm (Red-Red-Red)			
	Datasheet:				
	Source:	Farnell	RS	Mouser	DigiKey
		9339302			Larger Volumes (contact oomlout)

..Part Summary (detail):. (5 of 5)

	<table><tr><td>ID:</td><td>RES-561</td><td>QTY</td><td>30</td></tr><tr><td>Manufacturer:</td><td colspan="3"></td></tr><tr><td>Name:</td><td colspan="3">560 Ohm Resistor</td></tr><tr><td>Description:</td><td colspan="3">Resistor ¼ watt 560 Ohm (Blue-Green-Brown)</td></tr><tr><td>Datasheet:</td><td colspan="3"></td></tr><tr><td>Source:</td><td colspan="3"></td></tr><tr><td></td><td><i>Farnell</i></td><td><i>RS</i></td><td><i>Mouser</i></td></tr><tr><td></td><td>9339590</td><td></td><td><i>DigiKey</i></td></tr><tr><td></td><td></td><td></td><td><i>Larger Volumes (contact oomlout)</i></td></tr></table>	ID:	RES-561	QTY	30	Manufacturer:				Name:	560 Ohm Resistor			Description:	Resistor ¼ watt 560 Ohm (Blue-Green-Brown)			Datasheet:				Source:					<i>Farnell</i>	<i>RS</i>	<i>Mouser</i>		9339590		<i>DigiKey</i>				<i>Larger Volumes (contact oomlout)</i>				
ID:	RES-561	QTY	30																																						
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Name:	560 Ohm Resistor																																								
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	9339590		<i>DigiKey</i>																																						
			<i>Larger Volumes (contact oomlout)</i>																																						
	<table><tr><td>ID:</td><td>SERV-04-A</td><td>QTY</td><td>1</td></tr><tr><td>Manufacturer:</td><td colspan="3"></td></tr><tr><td>Name:</td><td colspan="3">Servo</td></tr><tr><td>Description:</td><td colspan="3">Mini Servo (Spring RC SM2309-S)</td></tr><tr><td>Datasheet:</td><td colspan="3">http://www.oomlout.com/C3A-D</td></tr><tr><td>Source:</td><td colspan="3"></td></tr><tr><td></td><td><i>Farnell</i></td><td><i>RS</i></td><td><i>Mouser</i></td></tr><tr><td></td><td></td><td></td><td><i>DigiKey</i></td></tr><tr><td></td><td></td><td></td><td><i>Larger Volumes (contact oomlout)</i></td></tr><tr><td></td><td></td><td></td><td>MOQ: 500</td></tr></table>	ID:	SERV-04-A	QTY	1	Manufacturer:				Name:	Servo			Description:	Mini Servo (Spring RC SM2309-S)			Datasheet:	http://www.oomlout.com/C3A-D			Source:					<i>Farnell</i>	<i>RS</i>	<i>Mouser</i>				<i>DigiKey</i>				<i>Larger Volumes (contact oomlout)</i>				MOQ: 500
ID:	SERV-04-A	QTY	1																																						
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			<i>DigiKey</i>																																						
			<i>Larger Volumes (contact oomlout)</i>																																						
			MOQ: 500																																						
	<table><tr><td>ID:</td><td>USBC-01-C</td><td>QTY</td><td>1</td></tr><tr><td>Manufacturer:</td><td colspan="3"></td></tr><tr><td>Name:</td><td colspan="3">USB Cable</td></tr><tr><td>Description:</td><td colspan="3">USB Cable 2m A to B</td></tr><tr><td>Datasheet:</td><td colspan="3"></td></tr><tr><td>Source:</td><td colspan="3"></td></tr><tr><td></td><td><i>Farnell</i></td><td><i>RS</i></td><td><i>Mouser</i></td></tr><tr><td></td><td></td><td></td><td><i>DigiKey</i></td></tr><tr><td></td><td></td><td></td><td><i>Larger Volumes (contact oomlout)</i></td></tr></table>	ID:	USBC-01-C	QTY	1	Manufacturer:				Name:	USB Cable			Description:	USB Cable 2m A to B			Datasheet:				Source:					<i>Farnell</i>	<i>RS</i>	<i>Mouser</i>				<i>DigiKey</i>				<i>Larger Volumes (contact oomlout)</i>				
ID:	USBC-01-C	QTY	1																																						
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	<i>Farnell</i>	<i>RS</i>	<i>Mouser</i>																																						
			<i>DigiKey</i>																																						
			<i>Larger Volumes (contact oomlout)</i>																																						

.:FILE SUMMARY:.

Vendor Suffix

Oomlout	OOML
ABRA	ABRA
Adafruit	ADAF
SparkFun	SPAR
Solarbotics	SOLA
Robot-R-Us	ROBO

(XXXX – Vendor Suffix)

(## - Circuit # (1-11))

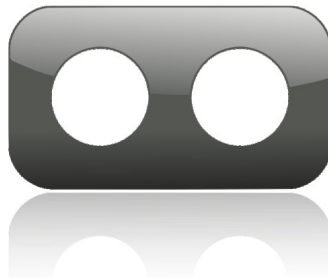
(%% - Guide page number (1-32))

Root Directory:

<http://ardx.org/src/>

/circ/	Files pertaining to the breadboard layout sheets
ARDX-CI-##.cdr	Breadboard layout source files (all vendors)
ARDX-SC-##.cdr	Schematic source file (all vendors)
ARDX-SC-xx-150.png	Schematic render (150 pixels wide png)
CIRC-00-sheet-XXXX.cdr	All 11 breadboard layout sheets in one file
CIRC-00-sheet-XXXX.pdf	All 11 breadboard layout sheets in one file (pdf)
CIRC-##-sheet-XXXX.cdr	Individual circuit breadboard layout
CIRC-##-sheet-XXXX.pdf	Individual circuit breadboard layout (pdf)
CIRC-##-sheet-.png	Individual circuit breadboard layout sheet (1500 pixels wide)
CIRC-##-3dexploded.skp	Circuit 3d model (exploded)
CIRC-##-3dexploded-XXXX.skp	Circuit 3d model (exploded) (some vendors)
CIRC-##-3dexploded.png	Circuit 3d Render (3000 pixels wide)
CIRC-##-3dexploded-XXXX.png	Circuit 3d Render (3000 pixels wide) (some vendors)
CIRC-##-3dexploded-454.png	Circuit 3d Render (454 pixels wide) (some vendors)
CIRC-##-assembled.jpg	Photo of assembled Circuit
CIRC-##-code.txt	Code for circuit
CIRC-##-square.jpg	Square detail photo
CIRC-##-square-100.jpg	Square detail photo (100 pixels)
CIRC-##-square-150.jpg	Square detail photo (150 pixels)
/guide/2/	Files pertaining to the guide
ARDX-EG-%%.cdr	Guide page source file
ARDX-EG-XXXX.cdr	Compiled Guide Source File
ARDX-EG-XXXX-WEB.pdf	Web quality guide (~1MB)
ARDX-EG-XXXX-DD.pdf	Document Distribution quality guide (~13MB)
ARDX-EG-XXXX-PRINT.pdf	PRINT quality guide (~34MB)
/holder/	Files pertaining to the acrylic holder
ARDH-01-laser.cdr	Holder source file
ARDH-01-laser.dxf	Holder file (dxf format)
ARDH-01-laser.ai	Holder file (adobe illustrator format)
ARDH-01-laser.svg	Holder file (scalable vector format)
/package/	Files pertaining to the packaging
ARDX-BX.cdr	Box front cover source file
ARDX-BX.pdf	Box Front (pdf)
ARDX-FM.cdr	Layout of parts to be packed in foam
ARDX-FM.pdf	Layout of parts to be packed in foam (pdf)

www.oomlout.com



.:any questions please contact us:.
info@oomlout.com